MODULE 8

Java Project Structure and Modularization

Organize Java applications using a well-defined project structure and modular design for maintainability, scalability, and reuse.

TEKsystems Global Services

WORKFORCE DEVELOPMENT

TEKsystems Global Services

WORKFORCE DEVELOPMENT

MODULE 8 OVERVIEW

A well-structured Java project is the foundation of a successful, maintainable enterprise application.

- Large enterprise applications may contain thousands of classes — without proper organization, code becomes difficult to maintain, teams conflict during development, and testing becomes difficult.
- This module covers the standard Java/Maven project layout, layered architecture, package organization by responsibility, and how enterprise teams structure real Spring Boot applications.

DURATION & LAB

2 Hours Theory

Core concepts and architecture patterns.

45 Min or 3–4 Hr Lab

Timed path (starter) or full extended path.

Business Scenario

500+ developers, one banking platform.

KEY TAKEAWAY Structure is not just about folders — it's about organizing code by feature and responsibility.

WHY PROJECT STRUCTURE MATTERS

Today's structure means less complexity tomorrow — it improves code quality, productivity, and long-term maintainability.

UNORGANIZED PROJECT

- Files scattered everywhere
- Hard to find and understand code
- High coupling and tight dependencies
- Difficult to test and debug
- Hard to onboard new developers
- Changes are risky and time-consuming

VS

WELL-STRUCTURED PROJECT

- Easy to navigate and understand
- Clear separation of concerns
- Reduced coupling, higher cohesion
- Easier testing and debugging
- Faster onboarding for new team members
- Safer changes with lower risk

KEY TAKEAWAY A strong project structure is the backbone of a maintainable, scalable, and successful Java application.

LEARNING OBJECTIVES

By the end of this module, you will organize Java projects effectively and apply modularization principles.

1. Understand the standard Java project layout and its components.
2. Organize code using packages and follow naming conventions.
3. Apply modularization principles to achieve high cohesion and low coupling.
4. Structure code into layers and modular components.
5. Manage dependencies using build tools (Maven/Gradle) and configuration files.
6. Separate concerns using well-defined modules and packages.
7. Improve code reusability through modular design.
8. Enhance maintainability and scalability using a structured project.
9. Follow best practices for project structure and modularization.
10. Build a scalable Java application using a modular project structure.

WHAT YOU WILL GAIN

- **Better Code Organization**
- **Easier Navigation**
- **Improved Collaboration**
- **Faster Onboarding**
- **Higher Code Quality**

KEY TAKEAWAY Success outcome: design and structure Java applications that are well-organized, modular, and easy to extend.

ENTERPRISE SCENARIO: LARGE-SCALE BANKING APPLICATION

GlobalBank must support millions of customers, complex regulations, and constant change — with agility, reliability, and maintainability.

Millions
Customers Across Web, Mobile & Branch

1000s
Daily Transactions

100s
Business Capabilities & Workflows

Strict
Security, Audit & Regulatory Needs

SCENARIO OVERVIEW

- GlobalBank is a multinational bank with millions of customers and thousands of daily transactions.
- The application must support retail banking, loans, cards, payments, and compliance across multiple regions — with high availability, performance, and scalability.

BUSINESS DOMAINS

Retail Banking

Loans

Cards & Payments

Compliance

KEY TAKEAWAY A clear project structure and modular design turn complexity into manageable pieces.

LARGE SCALE BANKING APPLICATION

SAMPLE MODULAR STRUCTURE

MODULES OVERVIEW

Customer Management	Handles customer onboarding, profiles, KYC and AML.
Account Management	Manages all account related operations and lifecycle.
Transaction Processing	Processes all types of transactions and business rules.
Loan Management	End-to-end loan lifecycle management.
Payment Gateway	Integrates and processes digital payments securely.
Cards Management	Manages cards, limits and related transactions.
Investments & Wealth	Handles investment products and wealth management.
Notification	Sends alerts, notifications and communication.
Fraud Detection	Monitors and detects fraudulent activities in real-time.
Reporting & Analytics	Generates reports, dashboards and business insights.

DESIGN PRINCIPLES

- Modular & loosely coupled architecture
- High cohesion within modules
- Independent deployable services (optional)
- Scalable, secure and maintainable

This modular structure supports scalability, maintainability, security and enables teams to work independently.

SAMPLE MODULAR STRUCTURE

GlobalBank organizes its codebase into channel, business-capability, shared, and foundation modules.

CHANNELS

- Mobile App
- Web Portal
- Branch System
- API Gateway

BUSINESS CAPABILITY MODULES

- Customer Mgmt
- Accounts & Deposits
- Payments & Transfers
- Cards Mgmt
- Loans Mgmt
- Alerts & Notifications

SHARED / PLATFORM MODULES

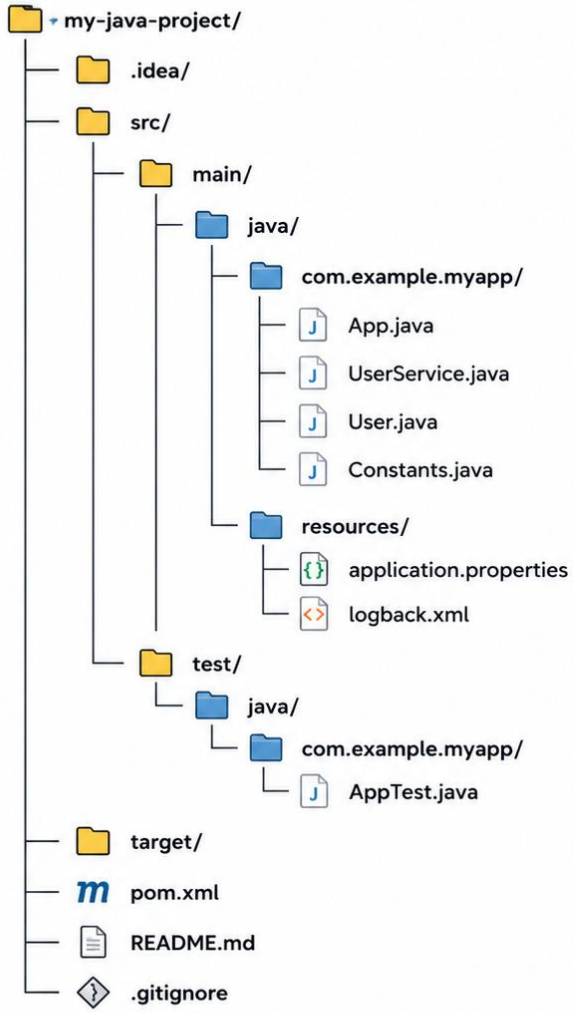
- Security & Auth
- Common Services
- Audit & Logging
- Reporting & Analytics
- Config Mgmt

FOUNDATION

- Database
- Messaging (Kafka)
- Cloud Infra
- CI/CD Pipeline

BUSINESS IMPACT Cuts technical debt and lets new teams or products (e.g., Digital Wallet) onboard with minimal impact.

JAVA PROJECT STRUCTURE



PATH	DESCRIPTION
.idea/	IDE (IntelliJ IDEA) configuration files
src/	Source code and resources
main/	Main application source and resources
java/	Java source code
com.example.myapp/	Base package for the application
App.java, UserService.java, User.java, Constants.java	Java classes (application logic, services, models, etc.)
resources/	Configuration and resource files
application.properties	Application properties (configurations)
logback.xml	Logging configuration
test/	Test source code
java/	Test Java source code
com.example.myapp/	Test package
AppTest.java	Unit test class
target/	Build output (compiled classes, JAR, etc.)
pom.xml	Maven project configuration
README.md	Project documentation
.gitignore	Git ignore configuration

NOTES

- This is a standard Maven-based Java project structure.
- Other build tools (Gradle, Ant) may have slight differences.

PURPOSE OF EACH DIRECTORY

Every top-level entry in the standard layout has one clear job.

DIRECTORY / FILE	PURPOSE
`src/main/java`	Main application Java source files (package structure)
`src/main/resources`	Configuration files, properties, static resources
`src/test/java`	Unit and integration test source code (mirrors main)
`pom.xml / build.gradle`	Manages dependencies, plugins, and build configuration
`README.md`	Project overview, setup, and usage instructions

KEY TAKEAWAY Follows a convention widely used in industry — promotes consistency across projects and teams.

STANDARD JAVA / MAVEN PROJECT STRUCTURE

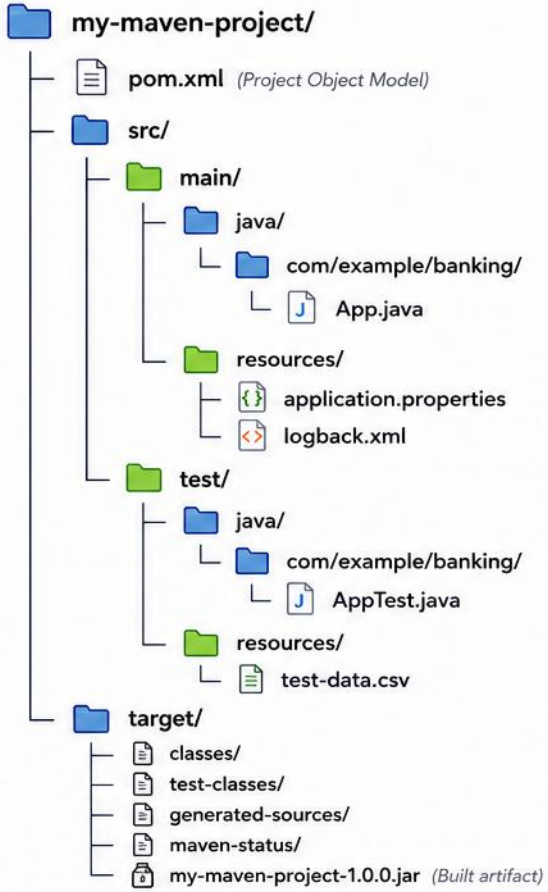
Maven enforces a standard project layout — convention over configuration — that makes builds predictable.

```
my-maven-project/
  src/main/java/                    // application source code
  src/main/resources/               // config, static files
  src/test/java/                    // unit + integration tests
  src/test/resources/               // test resource files
  pom.xml                           // POM: dependencies, plugins, build
  target/                           // build output (generated)
  .mvn/, mvnw, mvnw.cmd             // Maven Wrapper (optional)
  README.md

Example base package (src/main/java/com/example/app/): App.java, service/, repository/, model/, util/
```

KEY TAKEAWAY Maven uses convention over configuration. Lifecycle: validate → compile → test → package → verify → install → deploy.

MAVEN DIRECTORY LAYOUT

Standard Project Structure



DIRECTORY / FILE	DESCRIPTION
pom.xml	Project Object Model. Contains project configuration, dependencies, plugins, build settings.
src/main/java	Contains main Java source code. Follows package structure.
src/main/resources	Contains resource files used by the application (properties, XML, configuration, etc.).
src/test/java	Contains unit and integration test source code. Mirrors the main package structure.
src/test/resources	Contains resources for tests (test data, properties, config files).
target	Build output directory. Generated during build process. Contains compiled classes, test results, reports, and packaged artifacts.

Key Points

- Maven enforces a standard project structure.
- Keeps source code, resources, and tests organized.
- Build outputs are generated in the **target/** directory.

Benefits

- Convention over configuration
- Easy to maintain and scale
- Works seamlessly with Maven tools and IDEs (IntelliJ, Eclipse, VS Code)

A MINIMAL POM.XML EXAMPLE

The Project Object Model (POM) declares the project's coordinates, Java version, and how it builds.

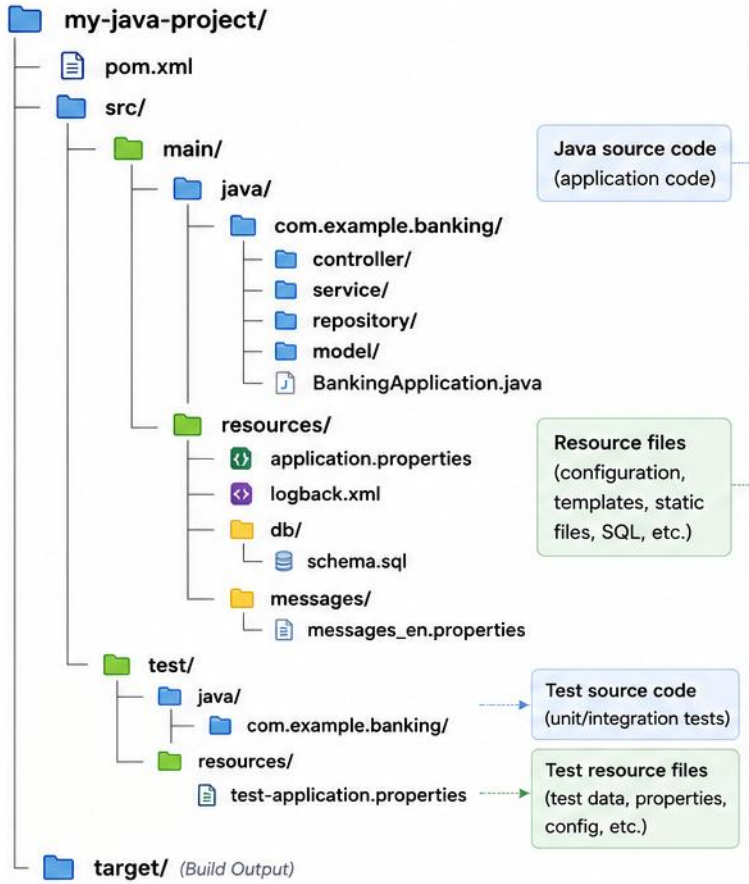
```xml
<project>
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example.banking</groupId>
  <artifactId>customer-service</artifactId>
  <version>1.0.0</version>
  <packaging>jar</packaging>

  <properties>
    <maven.compiler.release>21</maven.compiler.release>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>

  <dependencies>
    <!-- application dependencies go here -->
  </dependencies>
</project>
```

KEY TAKEAWAY pom.xml declares what the project IS (groupId/artifactId/version) and HOW it builds (Java release, dependencies) — Maven reads this file for every command.

Java Source and Resource Directories

Organizing Code and Resources in a Maven Project

Project Structure (Maven Standard)



1. Source Directories (Java Code)

Directory	Purpose
`src/main/java`	• Contains the main Java source code of the application. • Follows package structure (e.g., `com.example.banking`).
`src/test/java`	• Contains test source code. • Mirrors the main package structure for testing.

Only `.java` files belong in these directories.
They are compiled by Maven during the build process.

2. Resource Directories (Non-Java Files)

Directory	Purpose
`src/main/resources`	• Contains non-Java resources used by the application. • Examples: - `application.properties`, `logback.xml` - SQL scripts (`schema.sql`, `data.sql`) - Message bundles (`.properties`) - Static files, templates, certificates, etc.
`src/test/resources`	• Contains resources needed for tests. • Examples: test data, test properties, mock files, etc.

Resources are copied to the classpath during build.
They can be loaded using ClassLoader or Spring ResourceLoader.

Key Takeaway

Keep all Java code in the **source directories** (src/main/java, src/test/java) and all non-Java files in the **resource directories** (src/main/resources, src/test/resources).

Benefit:

This structure ensures clean organization, proper build, and easy maintenance.

TEST DIRECTORY STRUCTURE

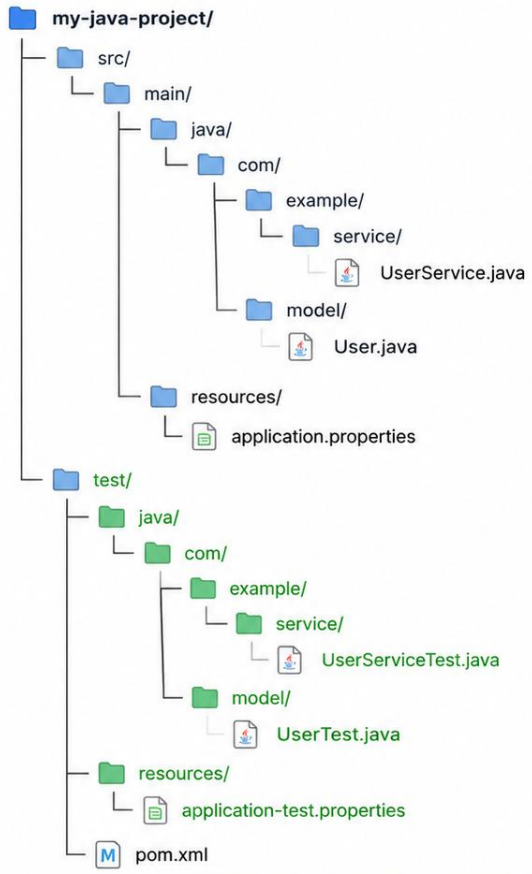
Test code lives under src/test/java, mirroring the same package structure as main code.

```
my-maven-project/src/
  main/java/, main/resources/
  test/java/com/example/banking/
    AccountServiceTest.java, AccountRepositoryTest.java
  test/resources/application-test.properties
```

TEST TYPE	LOCATION	EXAMPLE	PURPOSE
Unit Tests	src/test/java	AccountServiceTest.java	Test individual methods/classes in isolation
Integration Tests	src/test/java	AccountRepositoryTest.java	Test interaction between components/layers
End-to-End Tests	src/test/java	MoneyTransferE2ETest.java	Test the complete business workflow
Test Resources	src/test/resources	application-test.properties	Provide test data, configs, and utilities

KEY TAKEAWAY A well-organized test structure makes tests easy to find, run, and maintain. Keep tests fast, independent, and repeatable.

Java Project – Test Directory Structure



Legend

- Main source directory
- Test source directory
- Java source file
- Resource/config file
- Maven build file

Key Points

- All test code is under src/test/java
- Mirrors the package structure of main source (src/main/java)
- Test resources go under src/test/resources
- Allows Maven/Gradle to recognize and run tests automatically

Example Test Class: UserServiceTest.java

```java
package com.example.service;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter-api.Assertions.*;

class UserServiceTest {
    @Test
    void shouldCreateUser() {
        UserService service = new UserService();
        assertNotNull(service.createUser("John"));
    }

}
```

TRY IT YOURSELF — EXERCISE 1: READ A MAVEN LAYOUT

Reinforces: the Maven directory layout — src/main, src/test, docs, target — you just saw.

STEP	WHAT TO DO
Goal	Explain where production code, tests, config, docs, and generated files belong
File	maven-layout-notes.md (in examples/module-08-exercises/)
Classify	Six real files -> src/main/java, src/test/java, src/main/resources, docs/, or target/
Explain	target/ is generated by Maven — delete and rebuild anytime, so it is .gitignored
Deliverable	Six files correctly classified, target/ explained, no secrets committed in resources
Reference	exercises/exercise-01-maven-layout.md

```
$ tree customer-management-platform
customer-management-platform/
  pom.xml, docs/CODING-STANDARDS.md   // build id + team docs
  src/main/java/, src/test/java/   // production vs. test source
  target/   // generated by Maven — never commit, always .gitignored
```

TRY IT NOW Complete Exercise 1 before continuing — see exercises/exercise-01-maven-layout.md.

CHOOSING PACKAGE-BY-LAYER VS PACKAGE-BY-FEATURE

Both are valid choices for different contexts — pick based on project size, team structure, and domain complexity.

1. LAYER (ARCHITECTURAL) APPROACH

- Group by technical responsibility: controller, service, repository, domain.
- Groups related access together.
- Simple and familiar for smaller applications.
- Best for smaller applications and simple domains.

VS

2. FEATURE (DOMAIN-DRIVEN) APPROACH

- Group by business feature: customer, account, transaction.
- Each feature package has its own controller, service, repository, model, dto.
- Keeps packages cohesive and focused per feature.
- Best for larger systems, multiple teams, and complex domains.

3. HYBRID APPROACH

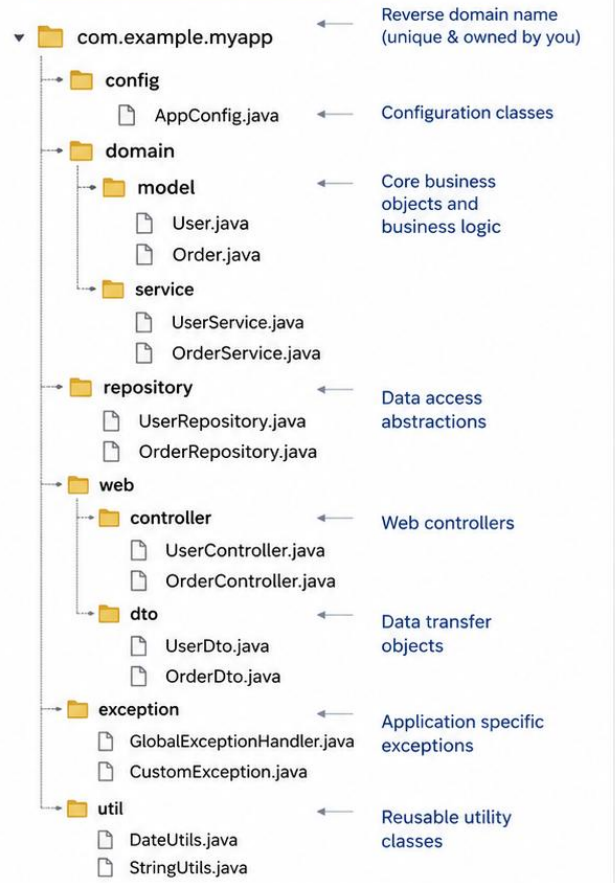
- A feature package containing its own controller/service/repository sub-packages — combines both approaches.
- Best practices: consistent, lowercase, reverse-domain package naming; avoid cyclic dependencies between packages.

KEY TAKEAWAY Layer, feature, and hybrid are all valid — choose based on project size, team structure, and domain complexity. Whichever you pick, keep package naming consistent and avoid cyclic dependencies.

Java Package Organization Best Practices

Well-structured packages improve maintainability, scalability and code clarity.

EXAMPLE: PACKAGE STRUCTURE



Keep packages focused and cohesive.
High cohesion, low coupling.

BEST PRACTICES

1. Use Reverse Domain Name

Start with your domain name in reverse order to ensure uniqueness (e.g., com.example.app).

2. Organize by Feature / Layer

Group related classes by feature or layer (e.g., controller, service, repository, model).

3. High Cohesion, Low Coupling

Keep related classes together.
Avoid putting unrelated classes in the same package.

4. Encapsulate Implementation

Keep internal classes and utilities in appropriate packages. Avoid exposing implementation details.

5. Keep It Simple and Flat

Avoid overly deep package nesting.
2–4 levels is usually enough.

6. Use Clear and Meaningful Names

Package names should be lowercase, concise, and meaningful. Avoid abbreviations unless well-known.

7. Consistent Across the Project

Follow the same structure and naming conventions throughout the project.

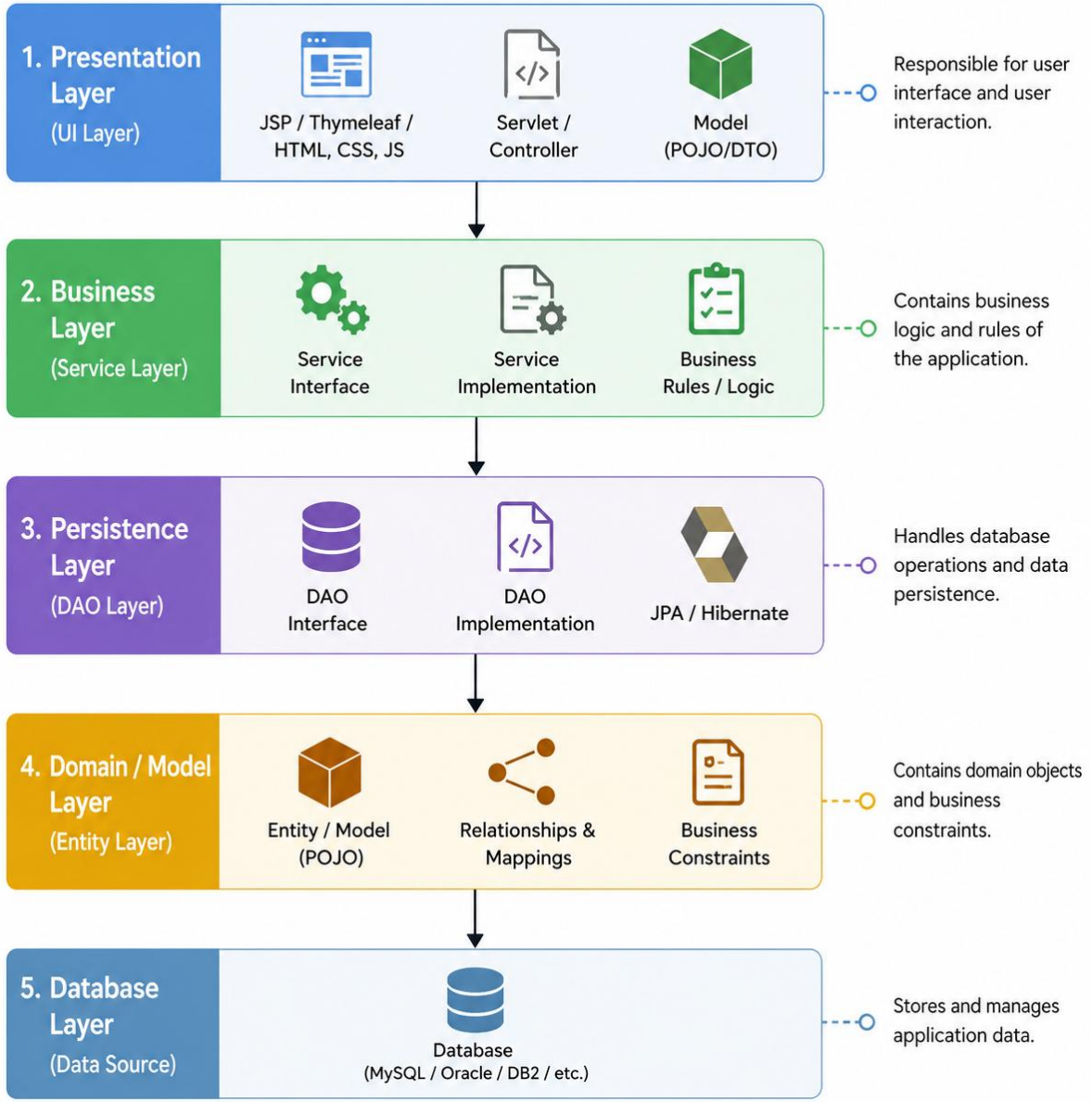
8. Put Utilities in the Right Place

Keep general utilities in a util (or common) package. Don't scatter them across features.

EXTRA TIPS

- Follow SOLID principles in package design.
- Refactor packages as the project evolves.
- Use IDE tools to enforce import and package conventions.
- Document package structure for new team members.

Layered Architecture in Java



LAYERED ARCHITECTURE — MINIMAL CODE SKELETON

Each layer only calls the layer directly below it — Controller depends on Service, Service depends on Repository.

```
// Repository layer -- data access only
public interface CustomerRepository {
    Customer findById(Long id);
}
// Service layer -- depends on Repository, not Controller
public class CustomerService {
    private final CustomerRepository repository;
    public CustomerService(CustomerRepository repository) {
        this.repository = repository;
    }
    public Customer getCustomer(Long id) {
        return repository.findById(id);
    }
}
// Controller layer -- depends on Service only
public class CustomerController {
    private final CustomerService service;
}
```

KEY TAKEAWAY Dependencies point one way: Controller → Service → Repository. Repository never imports Controller or Service types.

TRY IT YOURSELF — EXERCISE 2: PACKAGE ORGANIZATION

Reinforces: package-by-layer vs. package-by-feature and fully qualified naming you just covered.

STEP	WHAT TO DO
Goal	Map seven CRM types to packages and fully qualified class names
File	package-plan.md (in examples/module-08-exercises/)
Types	Seven types: Controller, Service, Repository, Customer, Request, AppConfig, exception class
Rule	Lowercase packages, reverse-domain root com.northstar.crm, folder path matches declaration
Deliverable	Seven FQCNs correct, DTO path matches declaration, packages lowercase and meaningful
Reference	exercises/exercise-02-package-plan.md

```
$ cat package-plan.md
com.northstar.crm.controller.CustomerController
com.northstar.crm.service.CustomerService
com.northstar.crm.dto.CustomerRequest  // src/main/java/com/northstar/crm/dto/...
com.northstar.crm.config.AppConfig  // reverse-domain root: com.northstar.crm
```

TRY IT NOW Complete Exercise 2 before continuing — see exercises/exercise-02-package-plan.md.

CROSS-CUTTING CONCERNS

Functionalities that affect multiple layers or modules — applied consistently across the whole application.

Security

- Authentication, authorization, encryption, input validation.

Logging

- Capturing application events for monitoring & troubleshooting.

Monitoring & Metrics

- Health checks and performance monitoring.

Exception Handling

- Centralized handling of exceptions and errors.

Caching

- Improves performance by reusing frequently accessed data.

Auditing

- Tracking user actions and changes for compliance.

Configuration Mgmt

- Managing application configuration across environments.

KEY TAKEAWAY Handle cross-cutting concerns centrally with AOP, filters/interceptors, or framework support (Spring, Jakarta EE).

CROSS-CUTTING CONCERN EXAMPLE: REQUEST LOGGING

A single Filter applies logging to every request, without touching controller or service code.

```
public class RequestLoggingFilter implements Filter {

    private static final Logger log =
        Logger.getLogger(RequestLoggingFilter.class.getName());

    @Override
    public void doFilter(ServletRequest req, ServletResponse res,
                         FilterChain chain) throws IOException, ServletException {
        long start = System.currentTimeMillis();
        try {
            chain.doFilter(req, res);
        } finally {
            long elapsed = System.currentTimeMillis() - start;
            log.info("Request handled in " + elapsed + "ms");
        }
    }
}
```

KEY TAKEAWAY Cross-cutting logic (logging, security, auditing) belongs in one shared component — not copy-pasted into every layer.

DTO PACKAGE STRUCTURE

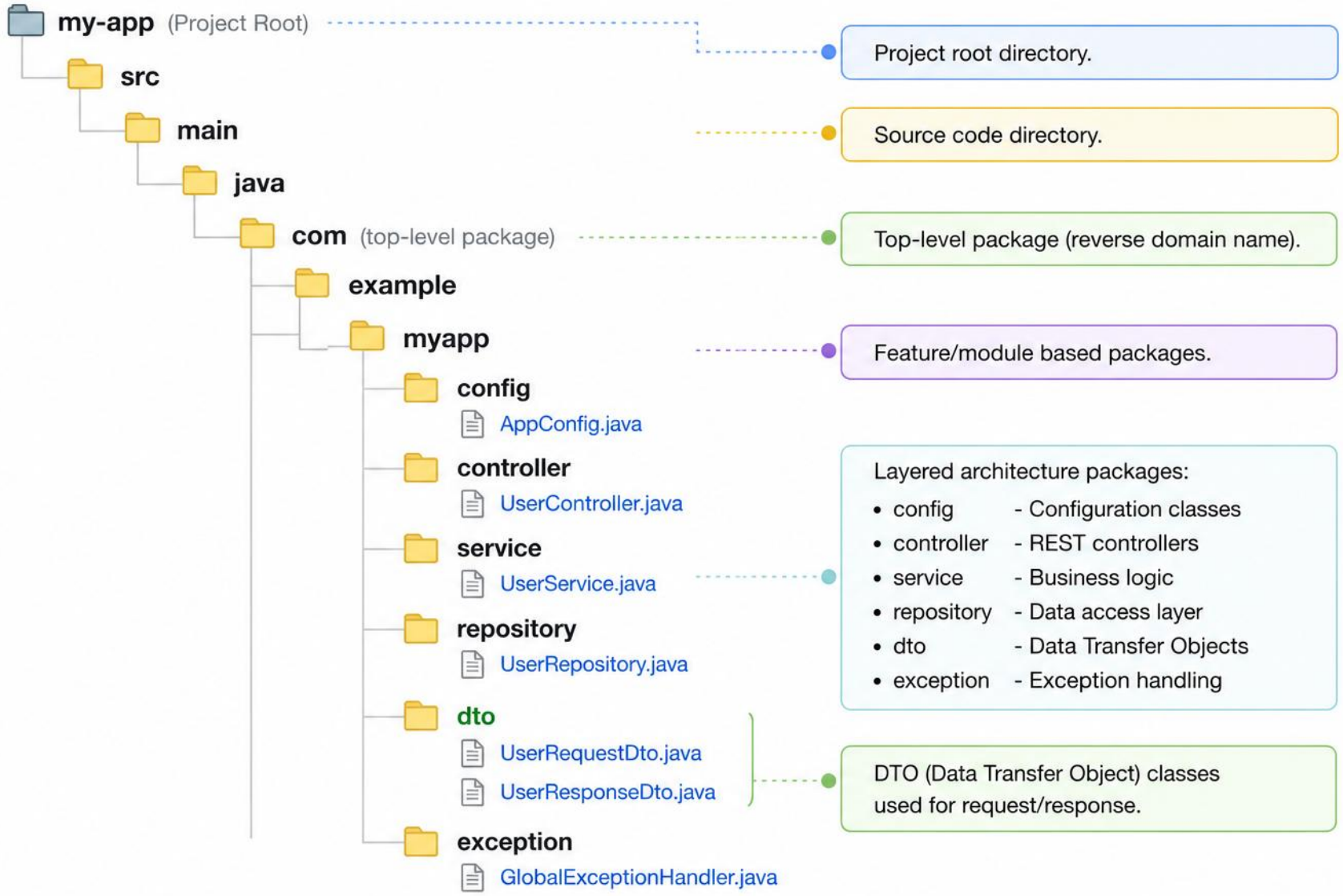
DTOs (Data Transfer Objects) move data between layers or across application boundaries without exposing entities.

```
com.example.project.dto/
  request/                    // CreateUserRequest.java, UpdateUserRequest.java
  response/                   // UserResponse.java, UserListResponse.java
  common/                     // AddressDTO.java (shared across modules)
  internal/                   // UserSummaryDTO.java, UserSearchDTO.java
  mapper/                     // UserMapper.java
```

GUIDELINE	WHY
Immutable, request/response separated	Do not expose internal entities directly

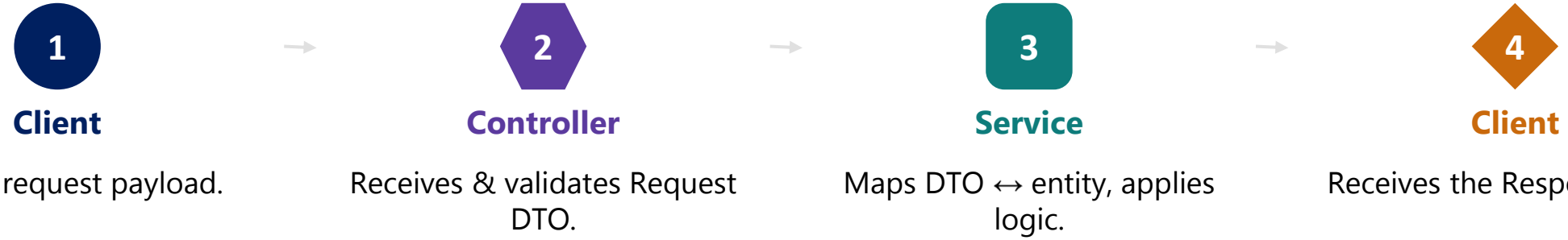
KEY TAKEAWAY DTOs decouple layers, expose only required data, and improve performance and security.

Java Package Structure Diagram



HOW DTOS FLOW THROUGH THE APPLICATION

Request DTOs enter at the Controller; Response DTOs leave the Service layer heading back out.



BEST PRACTICES

- Include only required fields; avoid business logic in DTOs; use meaningful and consistent naming.
- Validate request DTOs at the controller boundary using annotations (@NotNull, @Size, @Email).
- Use mapper frameworks (MapStruct, ModelMapper) or manual mapping for entity ↔ DTO conversion.

KEY TAKEAWAY A well-structured DTO package improves code organization, reusability, and maintainability.

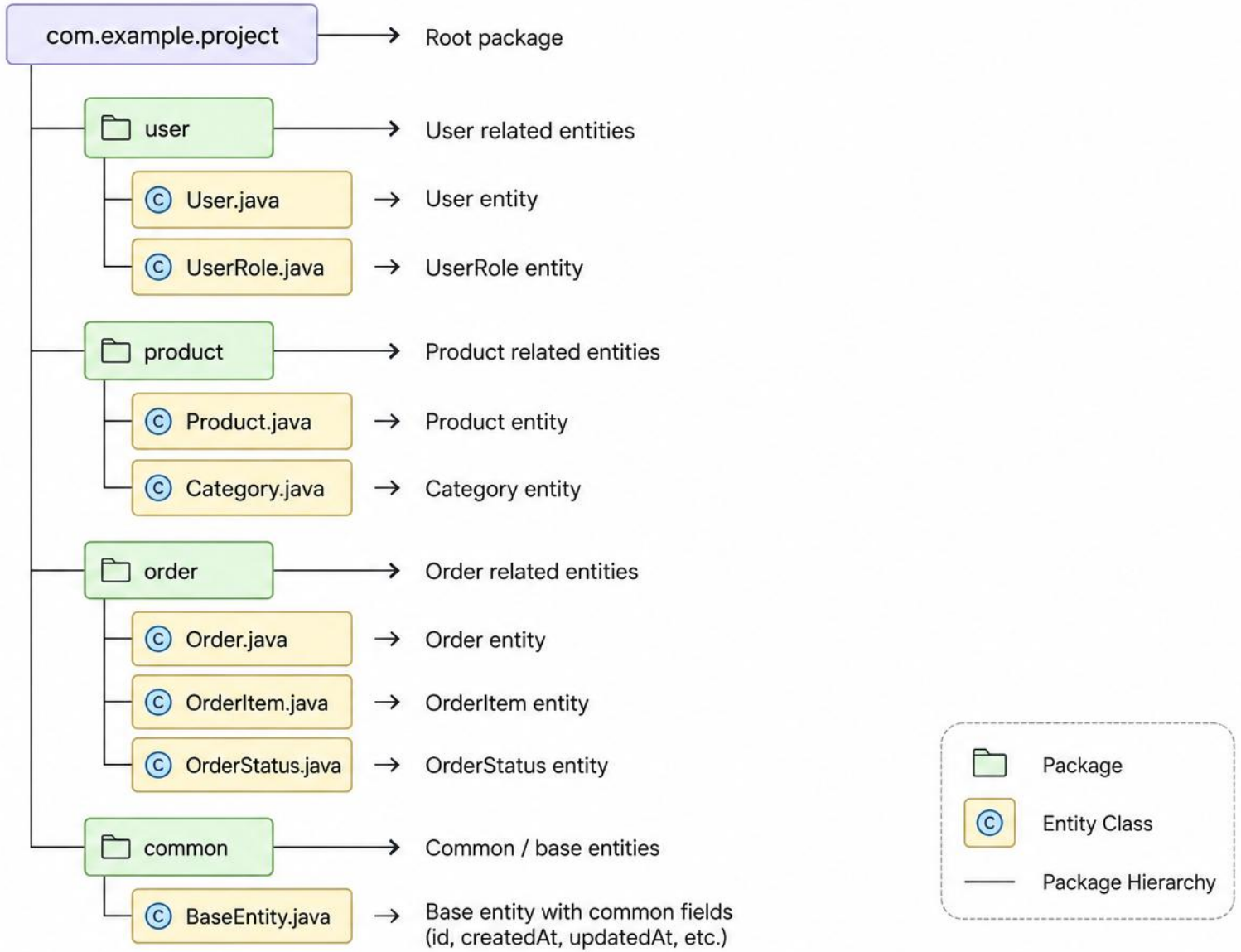
DTO CLASSES: REQUEST AND RESPONSE

Request DTOs validate input at the boundary; Response DTOs expose only what the client needs.

```java
public class CustomerRequest {
    @NotBlank(message = "Name is required")
    private String name;
    @Email(message = "Invalid email format")
    private String email;
    // getters and setters omitted for brevity
}

public class CustomerResponse {
    private String customerId;
    private String name;
    private String status;
    // getters and setters omitted for brevity
}
```

KEY TAKEAWAY CustomerRequest carries validated input in; CustomerResponse carries only safe, client-facing fields out.

Java Entity Package Structure



ENTITY DESIGN GUIDELINES

Keep orchestration and application workflow in services. Entities may contain focused domain behavior that protects their own invariants, but should not contain infrastructure or controller logic.

Map to Tables
Each entity maps to a single database table.

Use Relationships Properly
@OneToOne, @OneToMany, @ManyToMany.

Surrogate Key
Prefer a surrogate key (Long id) over business keys.

Auditing Fields
Use a base entity for createdBy, createdDate.

Encapsulation
Keep fields private, use getters/setters.

Protect Invariants
Focused domain behavior is fine; keep infrastructure/controller logic out.

ANNOTATIONS TO KNOW (INDUSTRY PREVIEW)

- @Entity, @Table, @Id, @GeneratedValue, @Column, @OneToMany/@ManyToOne, @Version — implemented when JPA is introduced in a later module.

KEY TAKEAWAY INDUSTRY PREVIEW: lazy loading and @Version optimistic locking are covered when JPA is introduced in a later module.

A JPA-STYLE ENTITY EXAMPLE

Entities map to tables and hold their own fields. (Industry Preview — JPA annotations are implemented in a later module.)

```java
@Entity
@Table(name = "customers")
public class Customer {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false)
    private String name;

    @OneToMany(mappedBy = "customer")
    private List<Account> accounts;

    protected Customer() {
    }
}
```

KEY TAKEAWAY @Id + @GeneratedValue give every entity a surrogate key; relationships like @OneToMany model how tables connect.

TRY IT YOURSELF — EXERCISE 3: ENTITY VS. DTO (STARTER)

Reinforces: entity vs. request/response DTO boundaries — a fill-in-the-blank starter, not from scratch.

STEP	WHAT TO DO
Goal	Prove a domain entity and boundary DTOs can have different fields and responsibilities
Starter	TODO / fill-in-the-blank skeleton — replace every ____ and // TODO; blanks do not compile
Files	mini-src/.../crm/: StructureDemo, entity/Customer, dto/CustomerRequest, dto/CustomerResponse
Entity vs. DTO	Customer (id/name/status) vs. Request (name/email) vs. Response (+summary())
Expected output	CUS-1001 | Amina Khan | ACTIVE
Reference	exercises/exercise-03-entity-vs-dto.md

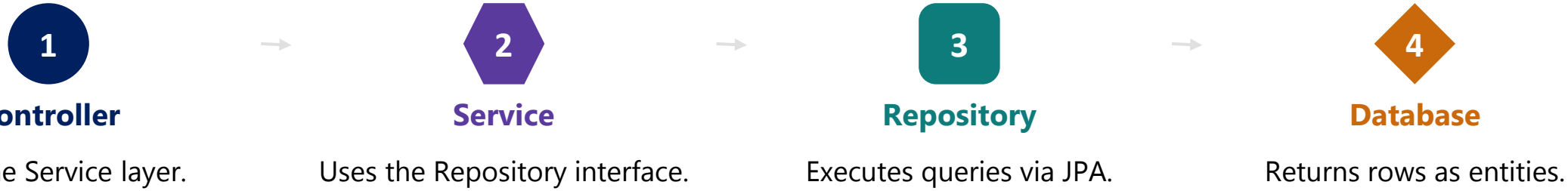
```
$ javac -d mini-out mini-src/com/northstar/crm/entity/Customer.java \
      mini-src/com/northstar/crm/dto/Customer*.java \
      mini-src/com/northstar/crm/StructureDemo.java
$ java -cp mini-out com.northstar.crm.StructureDemo
  // prints: CUS-1001 | Amina Khan | ACTIVE
```

TRY IT NOW Complete Exercise 3 (fill in every blank, then compile) — see exercises/exercise-03-entity-vs-dto.md.

Java Repository – Package Structure Diagram

REPOSITORY — HOW IT FITS & BEST PRACTICES

Prefer interfaces and let Spring Data JPA provide the implementation. (INDUSTRY PREVIEW — Spring Data JPA implementation starts in a later module.)



DESIGN GUIDELINES

- Use Interfaces — let Spring Data JPA provide implementations.
- Single Responsibility — one repository per aggregate/entity.
- Custom Queries — derived methods or @Query; keep logic minimal.
- Package Layout — group by feature: repository/user/, repository/product/, …
- Custom Implementation — separate impl package, *Impl suffix.
- Pagination & Sorting — use Pageable and Sort for efficiency.
- Specifications — use for dynamic, reusable queries.

BEST PRACTICES

- Name repositories as <Entity>Repository; extend JpaRepository or CrudRepository.
- Keep methods intention-revealing and focused; avoid business logic in repositories.
- Repositories participate in database operations, but application transaction boundaries are generally defined in the service layer (@Transactional); write unit tests for custom implementations.

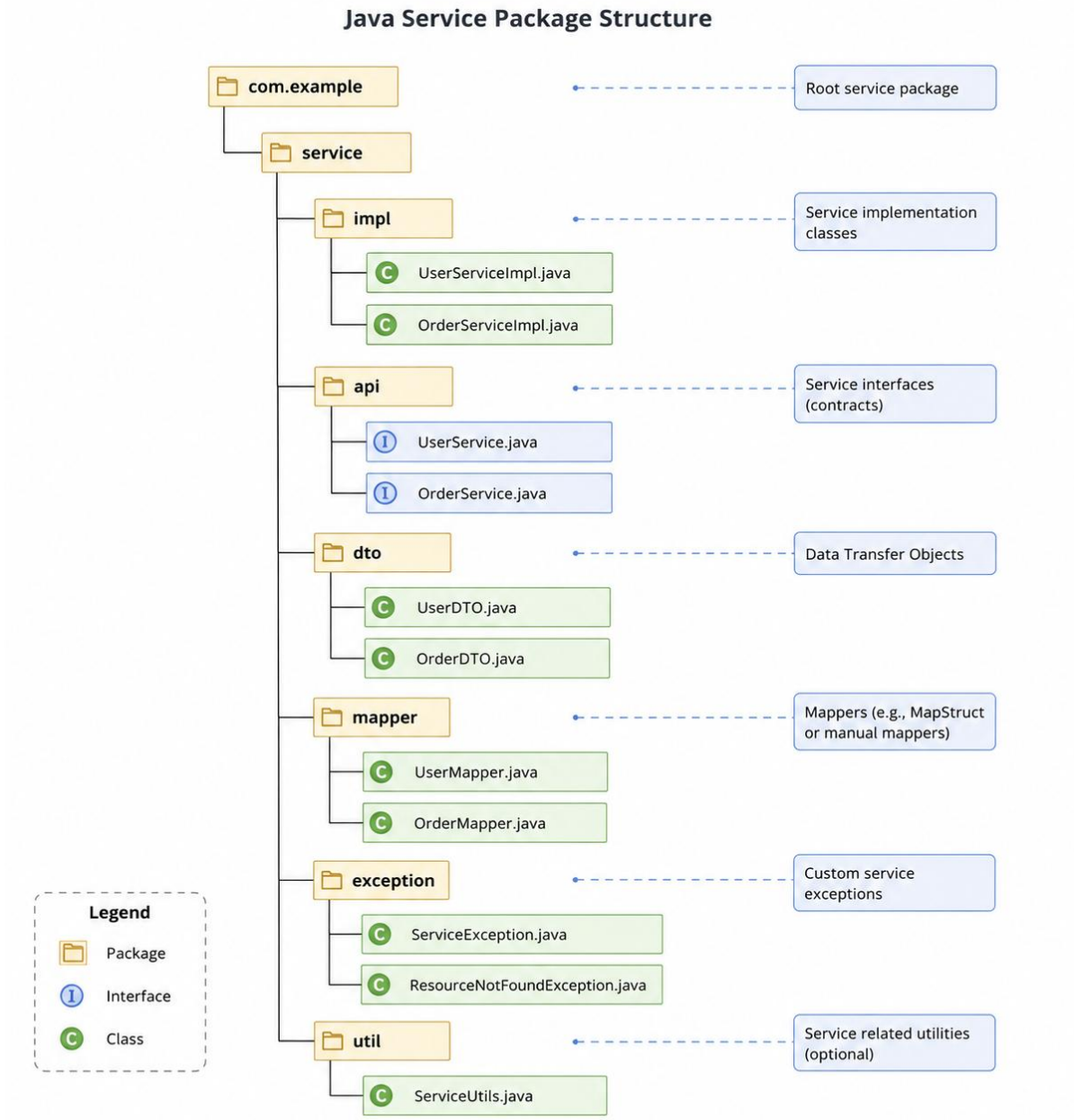
KEY TAKEAWAY A well-structured repository package ensures clean separation of data access logic.

REPOSITORY INTERFACE EXAMPLE

Spring Data JPA implements this interface at runtime — no implementation class to write. (Industry Preview.)

```
public interface CustomerRepository
        extends JpaRepository<Customer, Long> {

    Optional<Customer> findByEmail(String email);

    List<Customer> findByStatus(String status);

    @Query("SELECT c FROM Customer c WHERE c.name LIKE %:name%")
    List<Customer> searchByName(@Param("name") String name);
}
```

KEY TAKEAWAY Extending JpaRepository gives CRUD for free; derived methods and @Query cover everything else.



SERVICE DESIGN GUIDELINES

Each service should focus on a single business concern. Avoid circular or convenience-only service dependencies — use service-to-service collaboration for legitimate orchestration.

Single Responsibility

One service, one business concern.

Keep Stateless

Services should not hold request state.

Validate at Service Layer

Business validation before repository calls.

Use DTOs

Accept and return DTOs, not entities.

Handle Exceptions

Translate exceptions into business exceptions.

@Transactional

Manage transactions at the service layer. (Industry Preview.)

MORE BEST PRACTICES

- Name services by business domain (e.g., UserService); follow the package layout service/<feature>/ with <Entity>Service interface plus <Entity>ServiceImpl implementation.
- Log important business events and follow a consistent code style; write unit tests for every service.
- Use service-to-service collaboration when it represents legitimate business orchestration, and consider a dedicated orchestration service for complex workflows.

KEY TAKEAWAY A well-structured service package improves maintainability and makes the application easier to test and scale.

SERVICE INTERFACE AND IMPLEMENTATION

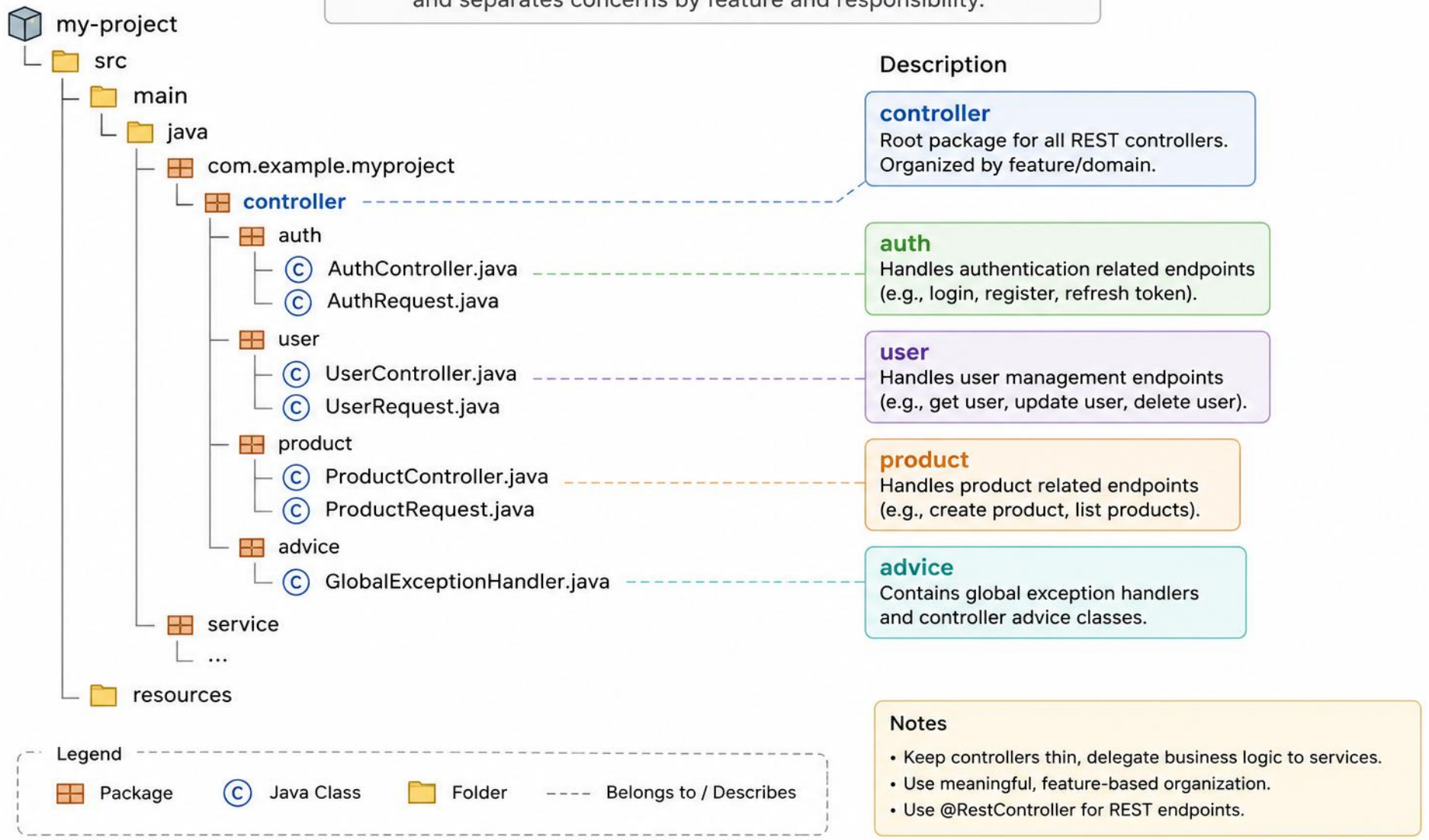
The interface is the contract; the Impl class holds business logic and orchestrates the Repository.

```
public interface CustomerService {
    CustomerResponse createCustomer(CustomerRequest request);
}
@Service
public class CustomerServiceImpl implements CustomerService {
    private final CustomerRepository repository;
    public CustomerServiceImpl(CustomerRepository repository) {
        this.repository = repository;
    }
    @Override
    @Transactional
    public CustomerResponse createCustomer(CustomerRequest request) {
        Customer saved = repository.save(new Customer(request.getName()));
        return new CustomerResponse(saved.getId(), saved.getName());
    }
}
```

KEY TAKEAWAY Controllers depend on the CustomerService interface, never on CustomerServiceImpl — the implementation can change freely.

Java Controller Package Structure

This structure follows a layered architecture (e.g., Spring Boot) and separates concerns by feature and responsibility.



CONTROLLER DESIGN GUIDELINES

Keep controllers thin — delegate all business logic to the service layer. (Spring-specific items below are INDUSTRY PREVIEW, implemented in later modules.)

Thin Controllers

Delegate business logic to services.

Request Validation

Use @Valid, @RequestParam annotations.

Meaningful Mappings

Consistent and RESTful endpoint mappings.

Use DTOs

Never return entities directly.

Secure Endpoints

Use Spring Security (@PreAuthorize).

Exception Handling

Use @ControllerAdvice globally.

MORE BEST PRACTICES

- Return consistent API responses with proper HTTP status codes; write slice tests (@WebMvcTest).
- Use appropriate HTTP methods, pagination/sorting/filtering for list APIs, and document with OpenAPI/Swagger.
- Follow the package layout controller/<feature>/ with <Entity>Controller RESTful endpoint mappings under /api/<resource>.
- Broader UI concerns (web pages/views, static resources) live outside this backend package and are handled by front-end tooling.

KEY TAKEAWAY A well-structured controller package ensures clean API design and improved maintainability.

REST CONTROLLER EXAMPLE

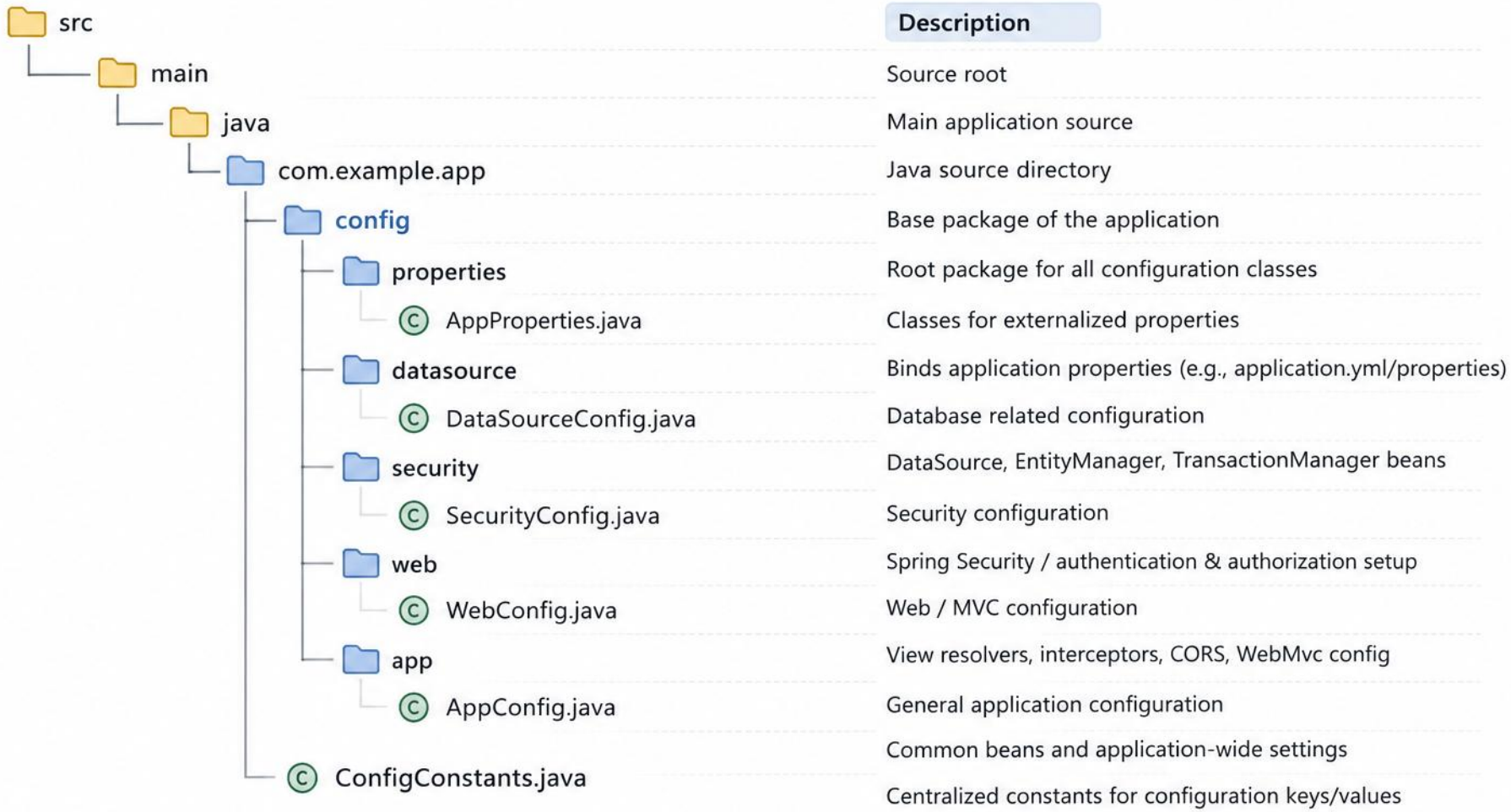
Controllers stay thin — validate input, delegate to the Service, and shape the HTTP response. (Industry Preview.)

```
@RestController
@RequestMapping("/api/customers")
public class CustomerController {
    private final CustomerService service;

    public CustomerController(CustomerService service) {
        this.service = service;
    }

    @PostMapping
    public ResponseEntity<CustomerResponse> create(
            @Valid @RequestBody CustomerRequest request) {
        return ResponseEntity.ok(service.createCustomer(request));
    }
}
```

KEY TAKEAWAY @Valid triggers the DTO's validation annotations; the controller never touches Customer, only DTOs.

Java Configuration Package Structure

Typical package structure for managing application configuration in Java



CONFIGURATION DESIGN GUIDELINES

Keep configuration classes clean and externalized — never mix business logic into config. (Spring-specific items below are INDUSTRY PREVIEW, implemented in later modules.)

Modular Configuration

Group config by concern: security, datasource, web.

Externalize Properties

Use application.yml, not hardcoded values.

Profile-Based Setup

Use @Profile for dev, test, and prod.

@Configuration & @Bean

Define and wire beans explicitly.

Enable Validation

Validate properties with @Validated.

Keep It Clean

No business logic — config only wires components.

MORE BEST PRACTICES

- Keep all configuration classes in a dedicated config package; use meaningful class and method names.
- Secure sensitive properties with environment variables or a secrets manager; document configs and write unit tests.

KEY TAKEAWAY A well-structured configuration package centralizes settings, promotes consistency, and makes the application easier to manage, test, and deploy across environments.

CONFIGURATION CLASS EXAMPLE

@Configuration classes wire beans explicitly — no business logic belongs here. (Industry Preview.)

```
@Configuration
public class AppConfig {

    @Bean
    public PasswordEncoder passwordEncoder() {
        return new BCryptPasswordEncoder();
    }

    @Bean
    @ConfigurationProperties(prefix = "app.datasource")
    public DataSourceProperties dataSourceProperties() {
        return new DataSourceProperties();
    }
}
```

KEY TAKEAWAY @Bean methods build objects the framework should manage; @ConfigurationProperties binds externalized settings to a typed class.

EXCEPTION PACKAGE STRUCTURE

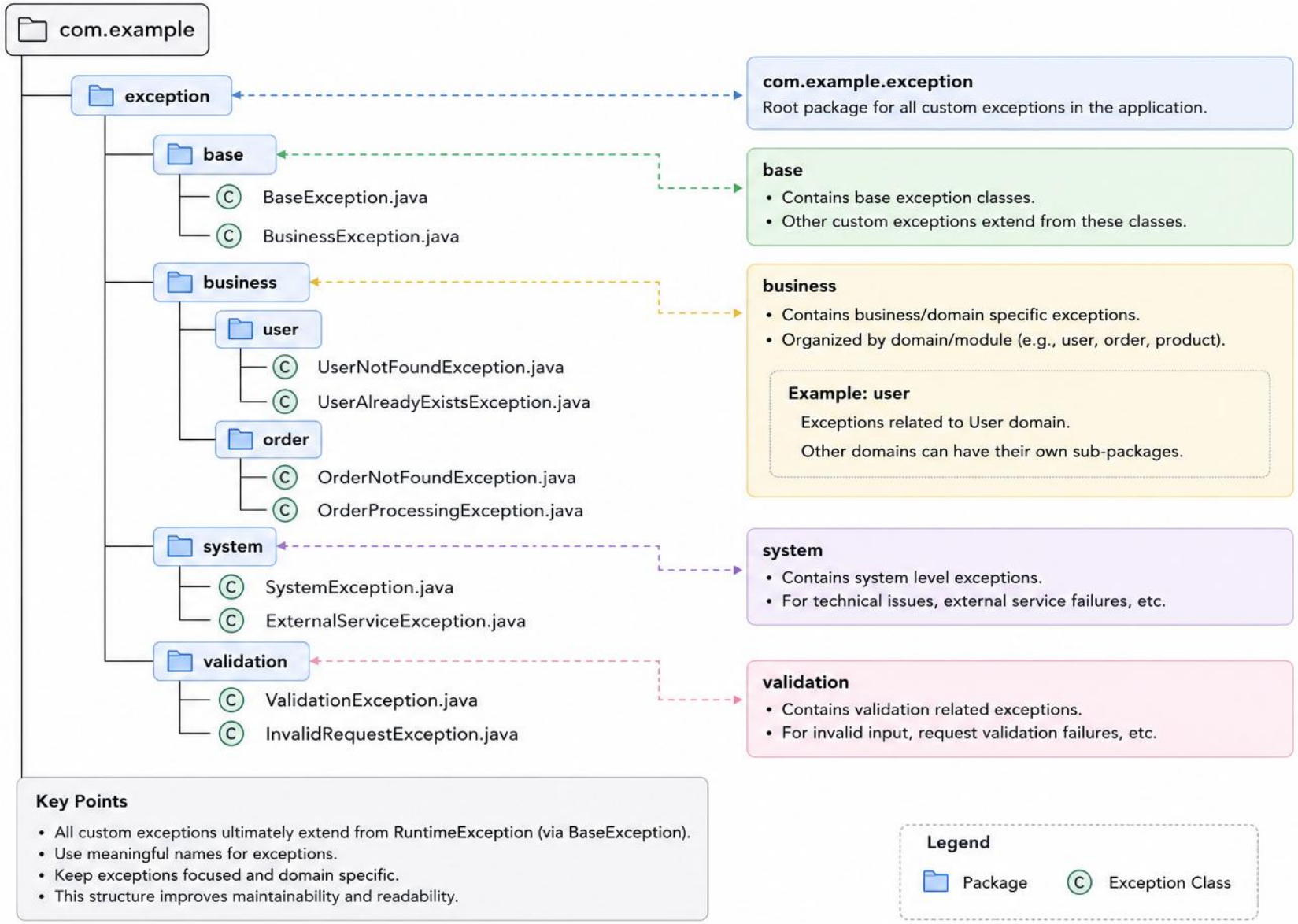
Custom exceptions used across the application, handled consistently through a global handler.

```
com.example.project.exception/
  base/                          // BaseException.java, BusinessException.java
  business/                      // ResourceNotFoundException.java, DuplicateResourceException.java
  handler/                       // GlobalExceptionHandler.java, ErrorResponse.java
  validation/                    // ConstraintViolationException.java
  security/                      // AuthenticationException.java, AuthorizationException.java
```

- Apply the Module 7 exception-handling standards (base hierarchy, specific exception types, never leak internals to clients) within this package structure.

KEY TAKEAWAY A well-structured exception package improves error handling — see Module 7 for detailed exception-handling standards.

Java Exception Package Structure



CUSTOM EXCEPTION AND GLOBAL HANDLER

A focused exception type paired with a centralized handler keeps error handling consistent everywhere.

```
public class CustomerNotFoundException extends RuntimeException {
    public CustomerNotFoundException(Long id) {
        super("Customer not found: " + id);
    }
}

@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(CustomerNotFoundException.class)
    public ResponseEntity<ErrorResponse> handle(
            CustomerNotFoundException ex) {
        ErrorResponse error = new ErrorResponse(ex.getMessage());
        return ResponseEntity.status(HttpStatus.NOT_FOUND).body(error);
    }
}
```

KEY TAKEAWAY Business code throws specific exceptions; one @RestControllerAdvice class turns every one of them into a clean HTTP response.

TRY IT YOURSELF — EXERCISE 4: LAYER RESPONSIBILITIES

Reinforces: the controller / service / repository / entity / dto / config / exception boundaries just covered.

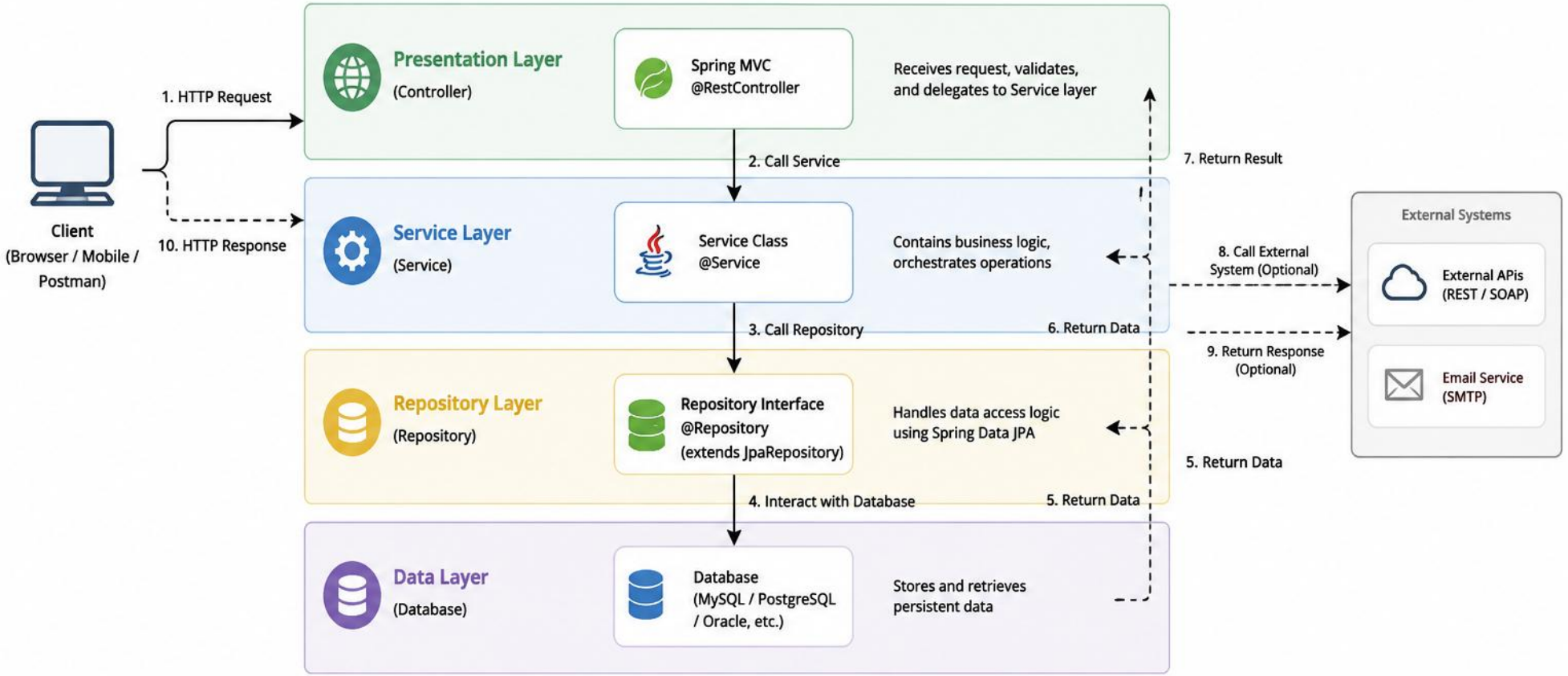
STEP	WHAT TO DO
Goal	Assign each CRM task to the layer that should own it
File	layer-responsibilities.md (in examples/module-08-exercises/)
Assign	Seven tasks: input handling, validation, find-by-ID, state, request shape, failure, wiring
Repair	Rewrite a "god controller" that validates rules, edits a list, and builds queries itself
Deliverable	Seven tasks assigned correctly, god-controller flow repaired, 2+ benefits explained
Reference	exercises/exercise-04-layer-responsibilities.md

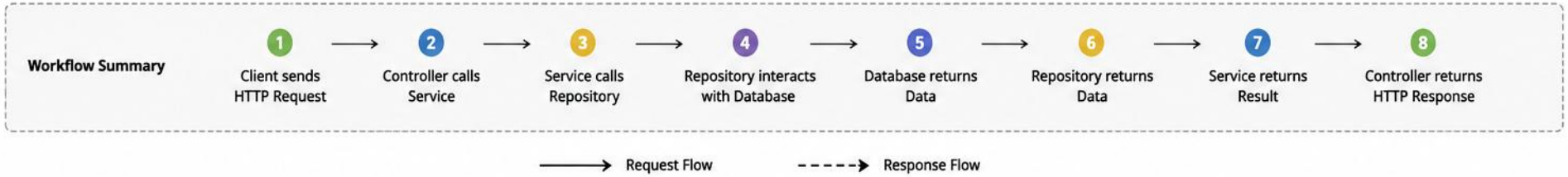
```
$ cat layer-responsibilities.md
accept input -> controller     validate -> service
find by ID -> repository       domain state -> entity
request shape -> dto           failure -> exception, wiring -> config
  // repaired: Controller -> Service -> Repository -> Service -> Controller
```

TRY IT NOW Complete Exercise 4 before continuing — see exercises/exercise-04-layer-responsibilities.md.

Layered Architecture Workflow (Java)

Request Flow from Client to Database and Back





DETAILED REQUEST FLOW

Eight steps take a request from the client, through every layer, and back again.

1. Client sends an HTTP request to the application.
2. Controller validates the request and converts input to DTO.
3. Controller calls the Service Layer, passing the DTO.
4. Service Layer contains business logic; may call multiple repositories.
5. Repository communicates with the database.
6. Data is fetched/updated and returned as Entities.
7. Service processes the data and returns DTO to Controller.
8. Controller sends the HTTP response back to the Client.

KEY TAKEAWAY Each layer can be tested in isolation using mocks — that's the payoff of a clean, layered request flow.

REQUEST FLOW, TRACED THROUGH REAL CODE

The same eight steps, expressed as an actual Controller -> Service -> Repository call chain.

```
@PostMapping                                    // 1-2. Client -> Controller
public ResponseEntity<CustomerResponse> create(
        @Valid @RequestBody CustomerRequest request) {
    return ResponseEntity.ok(
        service.createCustomer(request));       // 3. Controller -> Service
}

public CustomerResponse createCustomer(CustomerRequest request) {
    Customer entity = new Customer(request.getName());
    Customer saved = repository.save(entity);    // 4-6. Service -> Repo -> DB
    return new CustomerResponse(
        saved.getId(), saved.getName());        // 7-8. Service -> Controller
}
```

KEY TAKEAWAY Every arrow in the request-flow diagram is one method call in real code — nothing magic happens between the layers.

TRY IT YOURSELF — EXERCISE 5: TRACE A CUSTOMER REQUEST

Reinforces: the controller → service → repository request flow you just saw, success and failure.

STEP	WHAT TO DO
Goal	Document how a future create-customer request for CUS-1001 moves through the layers
File	customer-request-flow.md (in examples/module-08-exercises/)
Success flow	Client -> Controller -> Service (validate, assign ID/status) -> Repository -> Controller
Failure flow	A blank name stops at the Service layer — Repository is never called
Deliverable	Success + failure diagrams, DTO/entity transformations, a truthful now-vs-later table
Reference	exercises/exercise-05-request-flow.md

```
$ cat customer-request-flow.md
Name: Amina Khan   Email: amina@example.test
Requested status: ACTIVE   Correlation ID: lab-request-001
  // now: package names, stub responsibilities, documented flow
  // later: Spring annotations, JPA repository, HTTP response mapping
```

TRY IT NOW Complete Exercise 5 before continuing — see exercises/exercise-05-request-flow.md.

DEPENDENCY RULES AND BEST PRACTICES

Controller → Service → Repository: dependencies point one way only — lower layers never depend on upper ones.

KEY BENEFITS

Separation of Concerns

Each layer has one responsibility.

Maintain-ability

Changes in one layer stay local.

Testability

Each layer tested in isolation with mocks.

Flexibility

Tech changes in one layer don't affect others.

Reusability

Services and components reused across modules.

BEST PRACTICES

- Keep controllers thin — delegate all logic to services; use DTOs for data transfer between layers.
- Do not expose entities outside the service layer; use meaningful names for packages and classes.
- Handle exceptions and logging consistently across layers; write unit tests for each layer.
- Dependency direction (the rule, stated once): Controller → Service → Repository — lower layers (e.g., Repository) must never depend on controllers or web/HTTP concerns.
- Modularity: each layer is independently replaceable.

KEY TAKEAWAY Dependencies always flow Controller → Service → Repository — this keeps the architecture modular, testable, and maintainable.

TRY IT YOURSELF — EXERCISE 6: DEPENDENCY DIRECTION

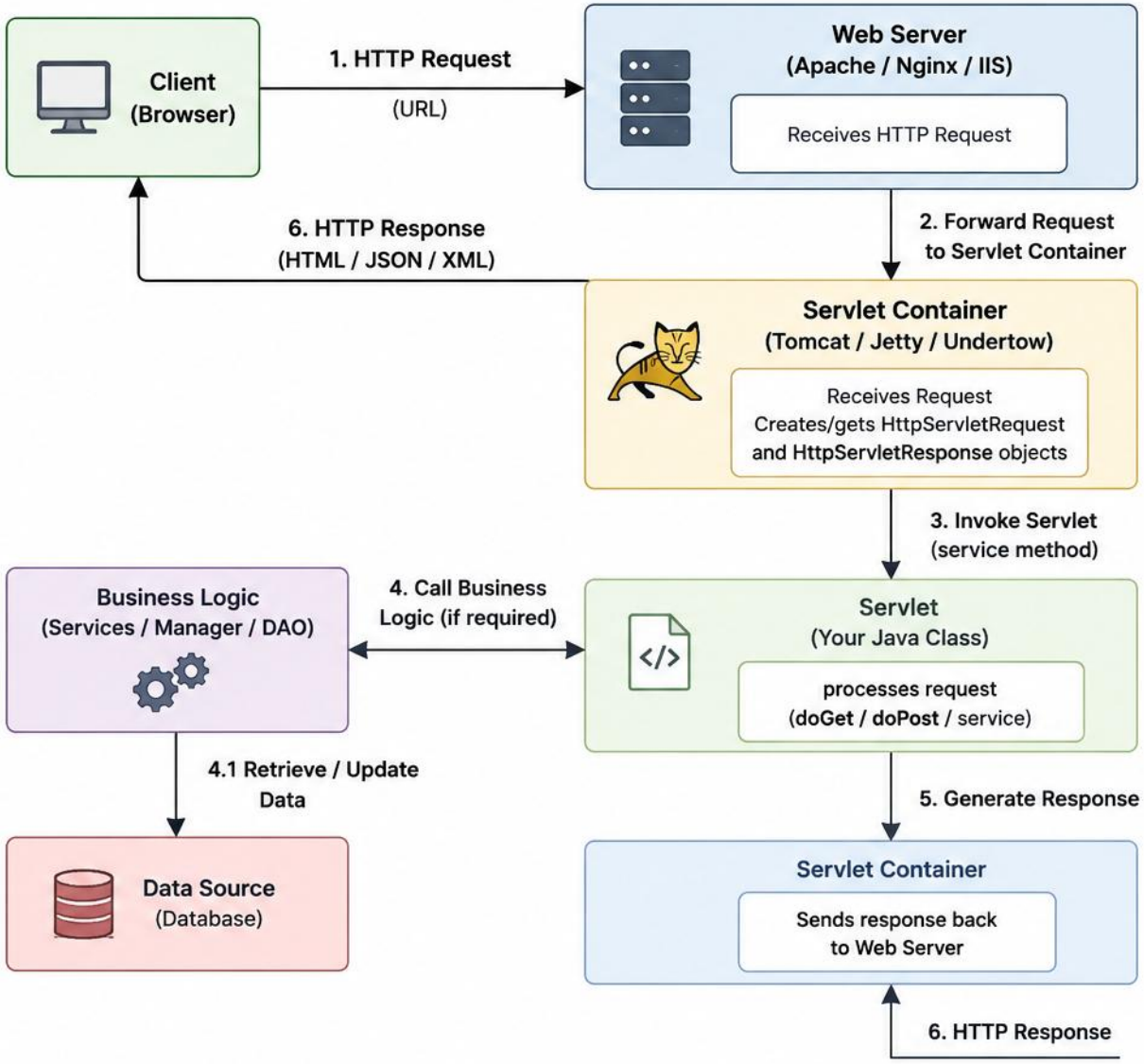
Reinforces: the Controller → Service → Repository dependency rule you just covered.

STEP	WHAT TO DO
Goal	Identify acceptable vs. problematic dependencies before they turn circular
File	architecture-rules.md (in examples/module-08-exercises/)
Classify	Seven dependencies across controller, service, repository, entity, dto — acceptable or not
Cycle	controller->service->repository->controller is a cycle; repair ends at ->entity
Deliverable	Seven dependencies classified, the cycle repaired, one written architecture rule
Reference	exercises/exercise-06-dependency-direction.md

```
$ cat architecture-rules.md
entity -> controller       // PROBLEMATIC: domain depends on transport
repository -> controller   // PROBLEMATIC: persistence depends on UI
service -> DTO              // needs context — avoid transport leakage
  // rule: entity/repository packages must never import controller classes
```

TRY IT NOW Complete Exercise 6 before continuing — see exercises/exercise-06-dependency-direction.md.

Java Request Processing Flow



1. Client sends an HTTP request to the server for a resource.
2. Web server forwards the request to the servlet container.
3. Servlet container creates request and response objects and invokes the appropriate servlet.
4. Servlet processes the request, may interact with business logic / services and prepare the response.
5. Servlet returns the response to the container.
6. Web server sends the HTTP response back to the client.

ENTERPRISE PROJECT EXAMPLE

Online Retail Order Management System — customers browse products, place orders, and track status.

KEY FEATURES

- **Registration & Auth**
- **Product Catalog**
- **Cart & Order Placement**
- **Payment Processing**
- **Order Tracking**
- **Admin Management**

TECHNOLOGY STACK

Java	MySQL	REST	JWT
Spring Boot	Data Store	API	Auth

KEY TAKEAWAY Domain: E-Commerce/Retail — build a scalable, secure, and maintainable Order Management System. A layered, well-organized backend also improves the customer experience — faster responses, fewer errors.

EXAMPLE FLOW: PLACE ORDER

One customer action, five layers of coordinated work.

1. **User browses & adds to cart**
2. **Controller validates, converts to DTO**
3. **Service checks stock, creates order**
4. **Repository saves order, updates inventory**
5. **Payment processed, confirmation sent**

PACKAGE	RESPONSIBILITY	PACKAGE	RESPONSIBILITY
`com.retail.controller`	Handles HTTP requests	com.retail.dto	Data Transfer Objects
`com.retail.service`	Business logic & rules	com.retail.exception	Custom exceptions
`com.retail.repository`	Data access layer	com.retail.config	Configuration classes
`com.retail.entity`	JPA entities / models	com.retail.util	Utility classes & helpers

KEY TAKEAWAY This enterprise project shows how layered architecture delivers a robust, scalable, maintainable solution.

KEY MODULES & SAMPLE ENTITIES

The Order Management System is organized into five business-facing modules, backed by seven core entities.

KEY MODULES

Product Mgmt

Products, categories, inventory.

Order Mgmt

Create, update, cancel, track orders.

Payment Mgmt

Process payments and refunds.

User Mgmt

Customers, roles, permissions.

Reporting

Sales and order reports.

SAMPLE ENTITIES

ENTITY	DESCRIPTION	ENTITY	DESCRIPTION
`User`	Stores user information	Payment	Payment transaction details
`Product`	Stores product details	Inventory	Tracks stock availability
`Category`	Product categories	OrderItem	Items in an order
`Order`	Order header information		

LAB OVERVIEW — PROJECT STRUCTURE AND ORGANIZATION

Build the Northstar CRM Maven skeleton — plain Java packages and stub classes (no Spring Boot yet). Timed path ~45 min in class; full path 3–4 hrs for homework.

LAB OBJECTIVES

1. Create the standard Maven layout and seven layer packages.
2. Separate DTO (API contracts) from entity (domain model).
3. Exercises 1–6 (already completed throughout this module) prepared you for this — now trace CUS-1001 through the layers on paper.

LAB CONSTRAINTS

- Plain Java 21 + Maven only — no Spring Boot, JPA, or HTTP framework.
- Seven layer packages with stub classes: controller, service, repository, entity, dto, config, exception.
- Stub methods throw UnsupportedOperationException by design — proves compile-time correctness before behavior.
- Target architecture: apply the layered-architecture diagram and dependency rules covered earlier in this module.

KEY TAKEAWAY A compiled, documented, stub-only skeleton — no Spring, JPA, or HTTP required yet.

LAB TASKS AND DELIVERABLES

Build the plain-Java Northstar CRM skeleton, complete the tasks, and submit the required deliverables.

#	TASK	DESCRIPTION
1	Project Setup	Create lab8-crm; add pom.xml (JDK 21, maven-compiler-plugin) and .gitignore
2	Package Structure	Create controller, service, repository, entity, dto, config, exception (seven packages under com.northstar.crm)
3	Entity & DTO Stubs	Customer entity plus CustomerRequest/CustomerResponse DTOs — empty shells, no JPA
4	Repository Layer	CustomerRepository.findById()/save() throw UnsupportedOperationException on purpose
5	Service Layer	CustomerService constructor-injects the repository; create()/getById() throw for now
6	Controller & Main	CustomerController delegates to service; Main prints banner; mvn clean compile + run
7	Config & Exception Stubs	AppConfig placeholder; CustomerNotFoundException(customerId)
8	Docs & Failure Experiments	docs/layer-flow.md, docs/CODING-STANDARDS.md; run 3+ documented failure experiments

KEY TAKEAWAY These tasks prove a compile-ready, correctly layered skeleton — ready for Lab 9.

DELIVERABLES & EVALUATION CRITERIA

What to submit, and how the lab is scored.

DELIVERABLES

- Maven skeleton (7 packages + stubs + Main) · CODING-STANDARDS.md + layer-flow.md · BUILD SUCCESS compile evidence · 3+ failure experiments · project README.

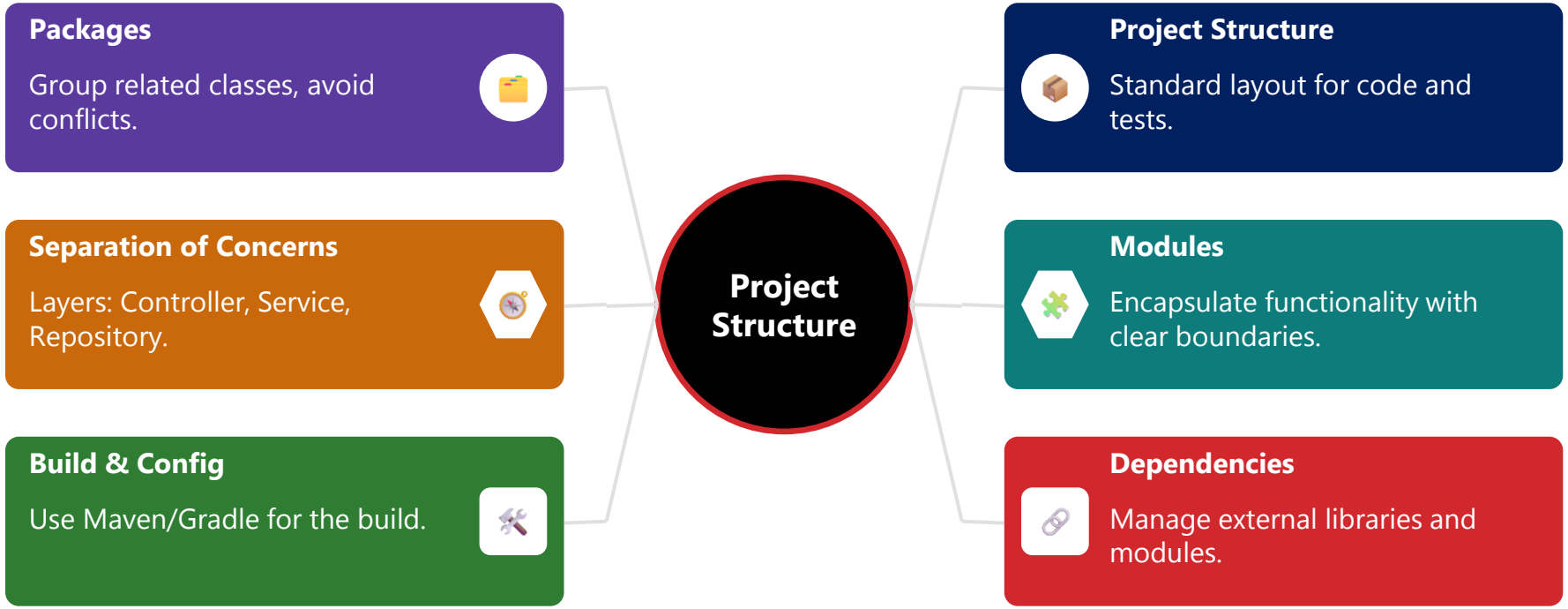
EVALUATION CRITERIA	WEIGHT
Core Implementation (packages, stubs, standards)	30%
Environment, Structure & Compile Correctness	25%
Failure Handling (stub failures / layer-rule experiment)	15%
Automated Verification (mvn compile, Main run)	10%
Security & Production Awareness	10%
Documentation & Evidence	10%

KEY TAKEAWAY Keep layers independent, prove compile success, and document decisions early.

MODULE 8 SUMMARY AND MODULARIZATION CHECKLIST

Organize Java applications using a well-defined project structure and modular design.

MODULARIZATION CHECKLIST High cohesion, low coupling, and encapsulation • Reusability and scalability • Standard layout (src/main/java, src/test/java) • Clear, reverse-domain package naming • Refactor structure regularly • Avoid cyclic dependencies • Faster onboarding and easier debugging.



KEY TAKEAWAY A well-structured and modular Java project — built on high cohesion, low coupling, and consistent naming — is easier to maintain, extend, and scale, leading to reliable software.

KNOWLEDGE CHECK (1 OF 3)

Test your understanding of the concepts covered in this module.

1. What is the primary benefit of layered architecture?

A. It increases code duplication

B. It tightly couples all components

C. Separation of concerns and maintainability

D. It makes the application slower

2. Which layer handles HTTP requests and responses?

A. Service Layer

B. Repository Layer

C. Controller Layer

D. Entity Layer

3. What is the purpose of DTOs?

A. To define database tables

B. To transfer data between layers

C. To handle exceptions

D. To configure the application

4. (Industry Preview) Which annotation handles exceptions globally in Spring Boot?

A. @Catch

B. @ExceptionHandler

C. @ControllerAdvice

D. @ResponseHandler

KEY TAKEAWAY Understanding each layer's role and proper package organization helps build robust, maintainable applications.

KNOWLEDGE CHECK (2 OF 3)

Four more questions to check your understanding.

5. (Industry Preview) Which Spring Data JPA interface provides CRUD operations and query methods?

A. CrudRepository

B. JpaRepository

C. EntityManager

D. JdbcTemplate

6. (Industry Preview) What is the purpose of the @Valid annotation?

A. To validate database entries

B. To validate incoming request data

C. To mark a class as a valid entity

D. To validate application properties

7. (Industry Preview) Which annotation marks a class as a REST controller?

A. @Service

B. @Component

C. @RestController

D. @Repository

8. Which layer is directly responsible for interacting with the database?

A. Service Layer

B. Controller Layer

C. Repository Layer

D. DTO Layer

KEY TAKEAWAY Spring Data JPA's JpaRepository extends CrudRepository with paging, sorting, and batch operations.

KNOWLEDGE CHECK (3 OF 3)

Two final questions, plus the full answer key.

9. What is the recommended way to pass data from Service Layer to Controller Layer?

A. Entity

B. DTO

C. Map

D. List

10. (Industry Preview) Which file is typically used to configure database connection and other app properties?

A. application.yml (or .properties)

B. web.xml

C. logback.xml

D. pom.xml

ANSWER KEY

- **1-C 2-C 3-B 4-C 5-B 6-B 7-C 8-C 9-B 10-A**

KEY TAKEAWAY Understanding the roles of each layer, proper package organization, and best practices helps in building robust, scalable, and maintainable Java applications.

"Structure today means less complexity tomorrow."

MODULE 8 COMPLETE

Java Project Structure and Modularization

You can now design and structure Java applications that are well-organized, modular, and easy to maintain and extend.